

NutriAI — Automated Diet Plan Builder

BAX-423 Big Data · UC Davis GSM · Spring 2026 · Larry Pastor

1. System Overview

NutriAI generates a personalised 7-day, 3-meal-a-day dietary plan in under 2 seconds. Given a user profile (diet type, allergens, clinical conditions, calorie target), the system retrieves semantically relevant meals from a 12,000-food USDA FoodData Central snapshot, applies probabilistic exclusion filters, enforces clinical rules, runs post-processing guarantees, and returns a complete plan well within the 60-second rubric requirement.

Core capabilities:

- 12,000-food offline USDA snapshot (Foundation + SR Legacy + Survey FNDDS) stored in SQLite
- Bloom Filter allergen/diet exclusion — $O(1)$ lookup, zero false negatives
- FAISS IVF approximate nearest-neighbour retrieval over TF-IDF/SVD embeddings
- Clinical condition rules for IBS (FODMAP), GERD, Type 2 Diabetes, Hypertension
- Post-processing nutrient guarantees (iron, B12, potassium, fibre)
- Streamlit UI with 4-tab layout, PDF/CSV export, nutrient gap analysis

2. Data Pipeline (Dimension 1)

A custom USDA API pipeline (**fetch_usda_data.py**) pulls Foundation, SR Legacy, and Survey (FNDDS) foods via the FoodData Central REST API. The pipeline deduplicates on *lowercase(description) + food_category*, backfills nutrients via the POST /foods batch endpoint, and persists as SQLite (**foods.db**) and CSV (**foods.csv**) for offline use. Monash University Low-FODMAP guidelines (monashfodmap.com) inform the IBS keyword blocklist; NIH DRI tables provide RDA targets.

Stage	Detail
Source	USDA FoodData Central API — Foundation + SR Legacy + Survey (FNDDS)
Foods fetched	12,000 items after deduplication (exceeds 10,000 rubric threshold)
Nutrients	Calories, protein, carbs, fat, fibre, iron, calcium, B12, vitamin D, zinc, sodium, potassium, magnesium, omega-3
Storage	SQLite (foods.db) + CSV (foods.csv) — fully offline after first fetch
Index	FAISS IVF + TF-IDF/SVD embeddings cached to disk (faiss.index, embeddings.npy, tfidf.pkl)

3. BAX-423 Big Data Techniques (Dimensions 2 & 3)

Technique 1 — Bloom Filter (Probabilistic Sketching)

A Bloom Filter (**BloomFilter** class, pipeline.py) is built at plan-generation time from all allergen, diet-exclusion, and clinical trigger keywords for the user. Each food candidate is checked in $O(1)$ time before any database lookup. Parameters: capacity=50,000, fp_rate=0.001 — zero false negatives guaranteed.

Technique 2 — FAISS IVF (Approximate Nearest-Neighbour Embeddings)

Food descriptions are encoded with TF-IDF vectorisation + TruncatedSVD (256 dims). An IVFFlat index (nlist=32) enables sub-millisecond semantic retrieval. Query expansion adds condition-specific terms (e.g. 'low-FODMAP', 'low glycaemic') to bias retrieval toward clinically appropriate foods.

Technique	Baseline	Optimised	Speedup
Bloom Filter	Python set scan	$O(1)$ hash probe	3–8×
FAISS IVF	Brute-force flat index	IVF probe (nprobe=4)	1.4×

4. Test Persona Results (Dimension 5)

Check	Priya IBS·Veg·Dairy-free	Ravi GERD·Non-veg·GF	Mei Diabetes·Vegan·Tree Nut	James HTN·Pescatarian·Soy -free
21 meals generated	✓	✓	✓	✓
Calories 80–120% target	✓	✓	✓	✓
Zero allergen violations	✓	✓	✓	✓
Diversity score $\geq$ threshold	✓	✓	✓	✓
Clinical filter (condition)	✓	✓	✓	✓
Diet type enforced	✓	✓	✓	✓
Micronutrient guarantee	✓	✓	✓	✓
Gen time < 60s	✓	✓	✓	✓
Score	8/8	8/8	7/7	8/8

5. Clinical Rules (Capability 1)

Condition	Exclusions	Nutrient Thresholds
IBS (FODMAP)	Onion, garlic, wheat, high-fructose, stone fruit, legumes (Monash list)	Iron $\geq 80\%$ RDA daily
GERD	Citrus, tomato, fried, caffeine, chocolate, spicy, peppermint	B12 $\geq 80\%$ RDA daily
Type 2 Diabetes	High-GI foods (GI>55 proxy): white rice, sugar, candy, instant cereal	Fibre $\geq 25\text{g/day}$; sugars $< 20\text{g}/100\text{g}$
Hypertension (DASH)	High-sodium processed foods; saturated fat sources	Sodium $\leq 1,500\text{mg/day}$; potassium $\geq 80\%$ RDA

6. Key Design Decisions

TF-IDF + SVD over sentence-transformers: sentence-transformers caused segfaults on Python 3.13/macOS. scikit-learn TF-IDF + TruncatedSVD provides stable 256-dim embeddings with no PyTorch dependency.

Offline USDA snapshot: 12,000 foods (Foundation + SR Legacy + Survey FNDDS) cached in SQLite. Grader can run entirely offline; API key only needed to regenerate the snapshot.

Post-processing nutrient guarantees: FAISS retrieval alone cannot guarantee micronutrient coverage. A second-pass injector replaces the lowest-nutrient slot with a targeted food (iron-rich for Priya, B12 for Ravi, potassium for James, fibre for Mei).

Bloom filter over pandas filtering: Bloom filter runs in $O(1)$ per food for allergen exclusion; benchmarked 3–8× faster than set-scan. Pandas safe-pool filter runs as a secondary hard guarantee.

Mixed household support: Per-meal diet overrides allow different diet types per meal slot (e.g. vegan breakfast + non-vegetarian dinner). Each slot uses a dedicated safe pool built from its own diet type, enforced via $O(1)$ `fdc_id` gate before any candidate is selected.

Religious & cultural constraints: Halal (no pork, no alcohol, no lard/gelatin), Kosher (no pork, no shellfish), and Hindu (no beef) profiles are supported via dedicated Bloom filter keyword lists, enforced identically to allergen exclusions — zero false negatives guaranteed.

Diversity by cosine distance: Greedy selection maximises mean cosine distance from already-selected foods. Category rotation (max 2 same-category meals/day) prevents monotony.

7. System Architecture (Dimension 5)

File	Role	Key components
fetch_usda_data.py	Data pipeline	USDA API fetch (Foundation + SR Legacy + FNDDS), nutrient backfill, SQLite/CSV write, deduplication
pipeline.py	Core engine	BloomFilter, build_faiss_index, NutriAIPipeline.generate(), clinical filters, nutrient guarantees, diversity scoring
app.py	Streamlit UI	Profile intake form, meal plan display, nutrient gap charts, exclusion log, benchmark tab, PDF/CSV export
data/foods.db	Food database	12,000 foods, 14 nutrients per food, ingredients text for allergen matching
data/faiss.index	FAISS IVF index	256-dim TF-IDF/SVD embeddings, IVFFlat nlist=32, cached for sub-second retrieval

8. Limitations & Future Work (Dimension 5)

Limitation	Impact	Future fix
No true GI database	GI filtering uses keyword proxy, not numeric GI scores	Integrate glycemindex.com lookup table
FNDDS nutrient gaps	~4,000 Survey foods have no calorie data (API timeout during backfill)	Retry backfill or use FNDDS CSV download
No meal structure	Each meal is a single food item, not a composed dish	Add recipe graph to combine complementary foods
Static serving sizes	Serving sizes are calorie-proportional estimates, not nutritionally optimised	Constrained optimisation per slot
No user feedback loop	Plan doesn't adapt to user preferences over time	RL bandit for preference learning (bonus feature)

9. Rubric Dimension Summary

Dimension	Points	What was delivered
1. Data Pipeline	10	12,000 foods from USDA API (Foundation + SR Legacy + FNDDS); deduplication on description+category; SQLite+CSV offline snapshot
2. Filtering & Ranking	20	Bloom filter allergen exclusion (zero false negatives); FAISS IVF semantic ranking; clinical rule enforcement; diversity scoring reported
3. Course Techniques	20	Bloom filter (sketching) benchmarked 3–8× vs set-scan; FAISS IVF (embeddings+ANN) benchmarked 1.4× vs flat index; both shown in Benchmarks tab
4. Hosting & Deployment	20	Live at nutriai-bax423.streamlit.app; Explain feature shows exact exclusion reason per food; vegetarian/vegan/pescatarian/non-veg + halal/kosher/hindu + mixed household (per-meal diet overrides); clean 4-tab UI
5. Brief & Demo	30	This document (4 pages); persona pass/fail table; BAX-423 technique justification; benchmark results; limitations table